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PAIN AND INJURIES

PHOTO: TERJE AAMODT



WHAT IS PAIN, REALLY?

'THE AMOUNT OF PAIN YOU EXPERIENCE DOES NOT NECESSARILY RELATE TO THE AMOUNT OF TISSUE DAMAGE YOU HAVE SUSTAINED.'

LORIMER MOSELEY

As I wrote in the introduction, when I was 16 years old, I experienced pain in both middle fingers for a long time. I woke up in pain, climbed with pain and felt it pretty much all the time over a period of about six months. No one could tell me what was causing it, and I didn't accept doctors' advice to take it easy or take a break from climbing altogether. After six months, two doctors and many terrible sessions, doctor number three explained to me that I had growth plate fractures in both fingers – *'like Schlatter's in the fingers'*, I remember him saying – and that I was going to be completely fine as long as I didn't climb for six weeks. (Osgood-Schlatter's disease is a common cause of knee pain in growing children.) Even though it was a brutal message for an ambitious young athlete to receive, I still remember that the pain in my fingers almost disappeared that day: finally, I had an answer.

But it's not like being given that answer healed the damage – the growth plates were just as broken – but my understanding of the symptoms had changed. And with a changed understanding, my pain also changed. To

quote Moseley again:

'ANYTHING THAT CHANGES YOUR BRAIN'S EVALUATION OF DANGER WILL CHANGE PAIN.'

We can consider the experience of pain in the same way as the experience of seeing: colours are picked up by sensors in the retina and sent as electrical signals to the brain, which, with the help of all available information, must make sense of these signals. I think we can all agree that our visual experiences are vulnerable to errors – that is, we don't always see what we thought we saw, we see things that aren't there, or we don't see things that are actually there. The quicker we must make sense of the information, and the more stressed we are, the more vulnerable we are to misinterpretation. Similarly, nerve fibres dedicated to registering elements such as load, speed, temperature, pressure, stretch and vibration pick up signals in our tissues and send them to the brain. The brain then makes an assessment, based on numerous factors, about the extent to which we need protection. The more factors that point in the direction of danger, the greater the need for protection – and pain has always been a good indicator of danger, thus ensuring our protection.

The pain response depends on how we interpret the information we have available, and that interpretation is based on three questions:

1. Am I safe?
2. Do I have information to predict the future?
3. Do I have information to influence the future?

The safer we consider the situation to be, the less the need for protection, and thus less pain. Pain is a protective mechanism and an experience, coloured by situations we find ourselves in, and can be experienced differently in different settings based on how we interpret a situation. Most of all, it is based on how dangerous a situation is for us. And all climbers who are faced with the risk of having to climb less, climb easier or – God forbid – stop for a period, will experience this as a situation that requires both protection and action.

It is important to understand that tissue damage is not necessary for the experience of pain. It's sufficient that there is a *possible* injury, and a situation that is experienced as dangerous or threatening for the person concerned. Pain is one of the most important survival mechanisms we have, but it is overprotective by nature and vulnerable to misinterpretation. Muscle soreness is a good example – it's not just a type of pain we live well with, it's actually a pain we take pride in and which gives us satisfaction. What distinguishes this soreness from other load-related ailments? When should we challenge and push on, and when should we end the session and possibly rest? These are experiences we build throughout our lives, and thus we also must accept being in pain for periods.

When we are faced with a situation that feels dangerous, in the form of having possible consequences such as a break from climbing and, perhaps, a reduced quality of life, pain is an appropriate response until we know more about the situation. The pain response is made up by both bodily signals and how we interpret these signals, based on our own knowledge, our experiences and memories, in what context it has happened, and what consequences it may have for us. It is important to understand this so that we can see the difference between pain and injury.

SENSITISATION

A smoke alarm in your house has a crucial job. It notifies you if there is smoke from a fire in your house so that you can save both your house and your life. It is equipped with sensors that detect smoke, and if the amount of smoke becomes large enough, the alarm will go off. Sometimes, though, the alarm can go off when you are cooking or when steam comes out of the bathroom after a long, hot shower. But, in the name of safety, one time too many is better, right? Better to be safe than sorry?

Or ...?

Unlike a smoke alarm, the human nervous system is adaptable, and can turn its sensitivity up and down. Sometimes it happens as a natural consequence of the body's condition – just think how it can hurt to touch your skin when you have a fever. Sometimes it happens as a natural consequence of the situation you find yourself in – just think how vigilant you might feel if you were to walk alone at night through a park where you knew others had been robbed. This up-regulation of sensitivity has obvious advantages for our survival, and

therefore the same up-regulation will occur in response to injuries so that we don't worsen the condition. The more sensitive our warning system, the less of a stimulus required to trigger a response – in this case, pain. But what if it is not dangerous to load that body part after all? That it is rather the opposite, that we *must* load the tissues to rehabilitate an injury as best as possible? Put another way: if you like to light candles at home, it is not desirable to have a smoke alarm that goes off every time you light a match. Even though the smoke alarm has an absolutely crucial job, we must do something about the sensitivity of the system.

Sensitisation of the nervous system occurs after all types of injuries and ailments, and just being aware that pain does not always mean injury, and that our warning system naturally becomes more sensitive after an injury, can make us feel a little more confident that we can still exercise, train and climb. To get there, we need to expose ourselves to and form new positive experiences.

MOVEMENT OPTIMISM

From an evolutionary perspective, we have benefited from placing more emphasis on the negative than the positive. Avoiding something which has a potentially negative outcome gives us a greater chance of survival than seeking out the positive. Negative information is therefore weighted more heavily than positive, and this affects (among other things) attention, motivation, behaviour, physiology and decision making. This is called *negativity bias* and is certainly relevant when it comes to pain and injuries.

As previously described, the experience of pain is a protective mechanism, and when we assess the information available to us, we will to a greater extent emphasise the negative over the positive. This tendency can cause us to avoid painful movements and take a darker view of the future, which in turn leads to an even greater focus on the negative than on the positive. It is not without reason that a low degree of pain self-efficacy, catastrophising and depression are closely linked to long-term pain conditions, and it is absolutely essential that we reverse this trend in order to contribute to as short a period as possible, with the best possible outcome.

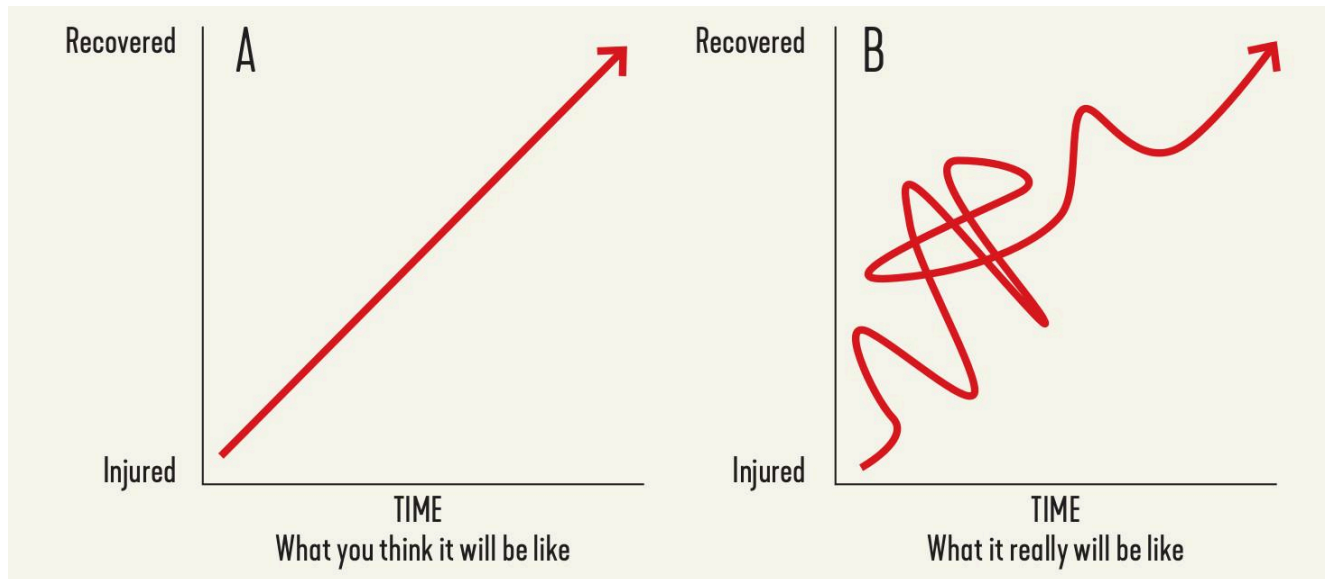
Optimism is defined as '*hopefulness and confidence about the future*', and an optimist expects positive things to occur in their future. This may sound naive, but optimism is closely linked to health-promoting effects – both physical and psychological. Optimism is also believed to influence our experience of pain and how we cope with painful conditions, and the outcome of a number of musculoskeletal disorders is closely linked to both positive expectations and a high degree of pain self-efficacy. Looking more brightly at the future creates more hope for recovery and more positive expectations for the course of the injury. This in turn increases the likelihood that we will complete the work that needs to be done, that we will better cope with fluctuating increases in symptom response, that we will manage relapses better and that we will return to our desired level sooner.

The same injury can therefore take two completely different courses in two different people who have different attitudes. A pulley injury can be seen as something catastrophic or something that is going to go well: as an injury which has ended the season and which creates uncertainty as to whether we will ever return to the same level; or as an opportunity to experience what this injury entails, an opportunity to work on other things in our climbing, and an opportunity to try other training methods which might even take us to a new level in the future. As Irish MMA fighter Conor McGregor said, '*Injury is not just a process of recovery, it's a process of discovery.*'

It is easier to be optimistic if we are given good information and advice that creates safety and predictability, and here comes a plea to healthcare professionals out there: we must familiarise ourselves with the types of injuries we encounter in practice, and with the sports and activities people take part in and want to return to. In this way, we can give good advice, emphasise the possibilities, and adapt the training to the individual's goals and wishes. We must be aware of the language we use, so that we do not create unnecessary fear and pessimism. Many of us are trained in a tradition where we must find faults and fix them, but we have been trained to a lesser extent in how to communicate such findings and the rehabilitation plans in a positive framework. If we can become as good and confident in this as in the job of diagnosing and making training plans, we can contribute to a far more optimistic recovery from injuries, and thus to a better prognosis and outcome.

No rehabilitation process has a linear progression, and it is important that healthcare professionals and unfortunate climbers are aware of this. Being prepared for the fact that the road can be bumpy and offer both stagnation and relapse creates a more realistic expectation than believing that rehab will progress in a linear fashion from day to day. The English physiotherapist Adam Meakins has created a humorous representation of

this, and I have this picture hanging in my office to help me to discuss the way forwards with those I meet at the clinic.



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Nor is it the case that we are either an optimist or a pessimist. Being optimistic is a quality that can be trained, just like anything else. It is a conscious choice as to how we want to look at the future. Injuries are a part of sport, unfortunately, and both reducing injury risk and managing injuries are performance factors in just the same way as training to get stronger fingers. We can train to be more optimistic, and such training programmes show promising results in working with cognitive strategies for long-term pain. By thinking about and writing down the desired outcomes where everything ends well, and writing down all the possibilities we have despite the injury, the focus shifts from the negative to the positive. Some days are of course shit: we're disappointed, down and sad, and this is understandable and acceptable. But we must then try to turn the tide and work our way back to a more optimistic view of things, as this is what opens up the door to a better end result.

A term that has been introduced in recent years is *movement optimism*. Based on knowledge of an injury, we can move away from the idea that pain is the same as injury and rather see the pain as a necessary part of the learning process in a rehabilitation course. Instead of avoiding movements we are afraid of or that cause pain, we can adjust the load and make the context safer so that we can expose ourselves to these movements. In this way, we can load the tissue, but also challenge the assumption that the movement is dangerous or causes an increase in pain.

EXPOSURE

We're exposed to things all the time, and we try to predict what will happen as a consequence of the things we are exposed to. If we have been exposed to something a lot, we will have formed experiences around it and so we can use these experiences to predict what will happen if we do one thing or another. When we are exposed to something we have not experienced before, we must form new experiences, and since the negative usually outweighs the positive, it will also shape what experiences we are left with and how we think similar situations will play out in the future.

A simple example of this is touching an electric fence. The first time you touch it, you learn that it hurts to get an electric shock. This experience means that you will (perhaps!) not touch an electric fence again, and thus avoid injuring yourself. Next, imagine that you are standing in front of another electric fence, but now you don't know whether it is turned on or not. How would you touch it if you had to? Of course, it depends on the situation, but in a safe environment most people would assume that they were going to get shocked, and thus be very careful about touching the fence. Your past experience means that a 50-50 situation is handled as something much, much worse, and you'd prefer to avoid a shock. Some may also have never touched an electric fence because they have heard that it hurts. They therefore don't need to have experienced something themselves to avoid a situation; it is sufficient to base their choices on the experiences of others. This also

applies in the event of an injury. If we have pain somewhere and then google the symptoms, we will try to predict the future based on other people's experiences. This is rarely good when it comes to being optimistic, since we will emphasise any negative information to a greater extent than any positive information.

Most of us have been sore after training. The first few times you experienced this you were perhaps unsure what was wrong, but were reassured by the surroundings and the people around you and were told that it wasn't dangerous and would pass within a few days. It did pass, and you have most certainly repeated this cycle numerous times throughout your life. Through this exposure to the symptoms of a low-grade muscle injury, we experience that it can be unpleasant and can inhibit the quality of training that day, but that it passes without major consequences. Exposure is therefore crucial for learning, and for us to be able to make good choices in the future.

I once asked an athlete I coached if she thought it was possible to be good at sports without experiencing periods of pain. She started laughing – she thought it was impossible. When I further asked her how she dealt with it, she said something that I believe is worth repeating:

'You must be able to put the various pains in boxes that you have control over. One box for sore skin, one for signs of illness, one for muscle soreness, one for aches and pains you can train with or around, one for aches and pains you need to pay attention to and investigate further, and one for the serious aches and pains that should take you out of training.'

EXPOSURE AND SYMPTOM RESPONSE

When we return to training and climbing after an injury, it is only natural that we experience some symptoms acutely and afterwards. But how much pain is okay? This is a difficult question to answer and it differs from injury to injury. With some injuries, such as a stress fracture in the growth plate in the finger of a young climber, we allow for little to no pain, while in other cases, such as an elbow tendinopathy, we allow as much pain as is acceptable for the individual. It is not a goal in itself that exercises and loading should hurt, but once we have ruled out serious injury, we can develop an acceptance of pain during and after exercise.

The most common measure of pain is the Visual Analogue Scale (VAS), where a patient rates their pain on a scale from 0 to 10 or 0 to 100, where 0 represents no pain. Remember that this is only a subjective experience of pain intensity, and that it is not necessarily the case that higher pain intensity equals greater damage. Also, remember that the experience of pain is affected by numerous factors, and that it can therefore be difficult to know exactly what led to something being better or worse today compared to yesterday or last week.

The advantage of using VAS is that it gives us the opportunity to see the relationship between load and symptom response over time. For example, the first sets of loading a finger after a pulley injury can cause symptoms, but then get better throughout the session and remain unchanged for the next 24 hours. We can then say that we've found a balance in the relationship between loading dose and symptom response.

A sore elbow may feel good after warming up, but then become significantly worse the following day and then persist for a few more days. Then it is likely that the loading dose was greater than the available capacity.

By following this relationship between load dose and symptom response over time, we can adjust the load up and down, and at the same time see that being able to do more without symptoms getting worse is also a big step in the right direction.

Day	1	2	3	4	5	6
What	Bouldering	Routes	Bouldering	Routes	Bouldering	Bouldering
	60 min	45 min	30 min	60 min	60 min	45 min
Intensity	Low	Low	Moderate	Low	Low	Moderate
Pain 0–10	Before: 2	Before: 2	Before: 2	Before: 2	Before: 1	Before: 1
	After: 4	After: 3	After: 4	After: 2	After: 2	After: 3

By using a training log, we can see what and how hard we have trained and monitor the pain response. It is common to measure the pain response on a scale from 0–10 where 0 equals no pain and 10 is the worst pain imaginable. Increased load usually results in an increased response, but if the pain returns to baseline during the following 24 hours, it is okay and we can continue the training. Gradually, we will be able to tolerate more load and see that the pain response decreases. On the other hand, in the event of a gradually increasing pain response, because of a too high training load, we can adjust the load down so that the trend doesn't become negative.



PHOTO: STIAN CHRISTOPHERSEN

Feeling a little bruised after a bouldering session outdoors is not so strange, given all the times we fall off during these sessions. These are symptoms we've become accustomed to and know will pass with a little movement and time. How do these symptoms differ from other symptoms? Is it just that we've become accustomed to them, while we're unsure what the burgeoning pain in the elbow means?

When we talk about how movement optimism and exposure affect pain, it is not to trivialise an injury or to claim that one should only think positively. But in many cases, we can probably intentionally dedramatise a good number of injuries and pain conditions. In this way, we can expose ourselves to movements that can be both scary and painful, so that we can create new experiences that, through successfully completing them, hopefully create optimism and faith that things will go well. A pulley injury, a hamstring injury or a painful elbow will naturally hurt when we load it. The million dollar question then becomes what that pain *means*. If we have control over the load that we expose the tissue to, we can at least state that we are not close to the forces that can damage the tissue. If we have control over the symptom response over time, we can be sure that we are balancing the load in a good way. And if we can work with movements that we're a bit scared of and preferably want to avoid, we can create new experiences and new predictions.

Maybe there was no electricity in the fence after all?